FreDA: Training-Free Test-Time Adaptation for Deepfake Detection via Non-Parametric Cache Retrieval

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Abstract. Deepfake (DF) detectors face significant challenges when deployed in real-world environments, particularly when encountering test samples deviated from training data through either distribution shifts or post-processing manipulations. To address these challenges, we propose FreDA (Free Detection Adaptation), a novel training-free test-time adaptation method that enhances the adaptability of detectors during inference without requiring any gradient computation or parameter updates. Our key idea is to leverage a non-parametric cache retrieval mechanism tailored to binary classification, enabling the model to dynamically incorporate distributional information from a small reference set via a similarity-based lookup and residual connection. We also introduce a fine-tuned variant, FreDA-F, which optionally refines cached keys with minimal training to further optimize performance under limited computational budgets. Empirically, our method demonstrates superior transferability and resilience against various perturbations across multiple benchmarks, including FaceForensics++, Celeb-DF-v1, and DFDCP. Compared to traditional fine-tuning paradigms, FreDA provides a parameter-efficient and highly adaptable framework that significantly enhances the generalization of deepfake detectors in diverse, real-world scenarios. Our code can be found at `https://github.com/123fangyang/FreDA`.

Keywords: Deepfake Detection · Test-Time Adaptation · Cache Model · Generalization · Training-Free.

1 Introduction

The rapid advancement of Generative Artificial Intelligence (GenAI) has enabled the creation of highly realistic deepfakes (DFs), posing severe threats to individual privacy and the integrity of public discourse [15]. High-profile incidents, including financial fraud through impersonation and politically motivated disinformation campaigns, underscore the urgency of developing robust deepfake detection systems [3,21]. In response, a wide range of detection approaches have been proposed, spanning spatial-based methods that analyze pixel-level artifacts

[16,1], frequency-based methods that exploit spectral signatures left by generative models [12,6], and hybrid approaches that unify both domains [22]. While these detectors have demonstrated promising performance under controlled evaluation settings, deploying them in real-world environments exposes two critical and unexpected challenges that severely undermine their reliability.

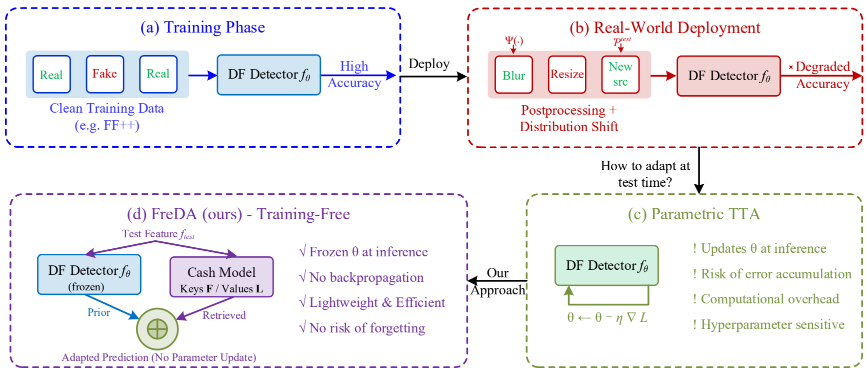

Fig. 1. Motivation for training-free test-time adaptation. (a) Detectors trained on clean data achieve high accuracy. (b) Postprocessing and distribution shift degrade performance at deployment. (c) Existing parametric TTA methods adapt by updating model parameters, introducing computational and stability risks. (d) Our proposed FreDA adapts via non-parametric cache retrieval without modifying model parameters.

The first challenge arises from *unknown postprocessing manipulations*. In practice, both authentic and manipulated media undergo various postprocessing operations, such as compression, resizing, blurring, and color adjustments, either incidentally through social media pipelines or deliberately by adversaries seeking to evade detection. Prior work [14,4] has shown that such operations can completely obscure the generation artifacts upon which detectors rely, particularly in the frequency domain. As the intensity of these manipulations increases, the spectral signatures that distinguish fake images from real ones become progressively indistinguishable, leading to significant performance degradation [19]. The second challenge stems from *distribution shift*: real-world test samples are frequently drawn from data distributions that deviate substantially from the training distribution, due to variations in generation methods, identity demographics, capture conditions, and image quality [20,9].

To mitigate these issues, several strategies have been explored, including employing data augmentation techniques during the training process [12,16], synthesis-based training [26], continual learning [20], and meta-learning [2]. However, these methods either require re-training the model or accessing the source data when facing the distribution shift at inference time. This requirement is impractical in many deployment scenarios due to privacy and storage constraints. More recently, Test-Time Adaptation (TTA) [28] has emerged as a principled framework for bridging the gap between training and deployment conditions. TTA methods adapt a pre-trained model to incoming test data at inference time without requiring access to the source training data or ground-truth labels, making them particularly well-suited to the dynamic and adversarial nature of DF detection in the wild. Nguyen-Le at al. [14] demonstrated the effectiveness of online TTA for DF detection through their T^2A method, which introduced uncertainty-aware negative learning to enhance detector resilience against both postprocessing manipulations and distribution shifts.

Despite its promise, T^2A method has a fundamental limitation: they require updating the model's parameters during inference through backpropagation. This parametric adaptation introduces several practical concerns. First, gradient-based updates incur computational overhead at test time, limiting deployment on resource-constrained devices and latency-sensitive applications. Second, iterative parameter updates carry the risk of catastrophic forgetting and error accumulation, particularly under non-stationary data streams where the test distribution may shift continuously [10]. These limitations motivate a natural question: *can we achieve the adaptability benefits of test-time adaptation for DF detection without modifying the model's parameters at inference time?*

In this paper, we answer this question by proposing **FreDA** (**Fr**ee **D**etection **A**daptation), a training-free test-time adaptation method for DF detection. Fig. 1 illustrates how our method is different from existing methods. Inspired by the non-parametric cache retrieval mechanism of Tip-Adapter [34], FreDA constructs a key-value cache model tailored to the binary classification structure of DF detection, where encoded features from a small reference set serve as keys and their corresponding labels serve as values. During inference, a test sample's feature representation is used as a query to retrieve relevant knowledge from the cache via similarity-based lookup, and the retrieved information is combined with the pre-trained detector's predictions through a residual connection. Crucially, the entire process involves no gradient computation or parameter update, preserving the original detector's learned representations while dynamically incorporating distributional information to handle domain shift and postprocessing corruptions. Furthermore, we introduce a fine-tuned variant, **FreDA-F**, which optionally refines the cached keys with minimal training epochs to further boost performance when limited computation is available.

Contributions. Our main contributions are summarized as follows:

- We propose FreDA, a novel training-free TTA method for DF detection based on non-parametric cache retrieval. To the best of our knowledge, this is the first work to achieve adaptation at inference time for DF detection without requiring any parameter updates at inference time.
- We conduct comprehensive experiments across multiple DF datasets under both in-domain and cross-dataset settings, as well as under various postpro-

cessing perturbations, demonstrating that FreDA enhances the performance of existing DF detectors at inference time against distribution shift.

2 Related Work

2.1 Deepfake Detection

DF detection methods can be broadly categorized into spatial-based, frequency-based, and generalization-oriented approaches. Spatial-based methods analyze pixel-level artifacts: Rössler et al. [25] established the FaceForensics++ benchmark using XceptionNet; subsequent works improved detection through consistency-driven representations [16], reconstruction-classification frameworks [1], localized artifact attention [13], and alternating spatial-temporal freezing for video [31]. Frequency-based methods exploit spectral signatures left by generative models, including GAN-specific frequency artifacts [6], phase spectrum analysis [12], and multi-scale frequency-aware mining via DCT decomposition [22].

A central challenge remains cross-dataset and cross-manipulation generalization. Shiohara and Yamasaki [26] proposed Self-Blended Images to encourage manipulation-agnostic representations; Pan et al. [20] tackled continual learning across expanding forgery types; and Chen et al. [2] introduced one-shot test-time training, though it still requires backpropagation during inference. More recently, vision-language models have emerged as powerful backbones: Ojha et al. [19] demonstrated that frozen CLIP features provide remarkably generalizable representations via linear probing, while Khan and Dang-Nguyen [8] showed that prompt tuning further improves performance by jointly adapting visual and textual components. These findings—that strong pre-trained representations combined with appropriate adaptation mechanisms can substantially improve robustness—directly motivate our cache-based approach.

2.2 Test-Time Adaptation for Vision Tasks

Test-time adaptation (TTA) addresses distribution shifts at deployment without requiring access to source training data or ground-truth labels [10]. Existing methods fall into two broad categories based on their adaptation mechanism.

Optimization-based TTA. The dominant paradigm relies on gradient-based parameter updates during inference. Representative methods include TTT [27], which adapts via a self-supervised auxiliary task; TENT [29], which minimizes prediction entropy by updating batch normalization layers; MEMO [33], which enforces consistency across augmented views; EATA [17], which selectively adapts on reliable samples with Fisher regularization; CoTTA [30], which handles continual shifts via stochastic weight restoration; and SAR [18], which stabilizes adaptation through sharpness-aware minimization. Despite strong results on corruption benchmarks, all these methods depend on backpropagation at inference time, introducing computational overhead, risks of catastrophic forgetting, and hyperparameter sensitivity [36].

Non-parametric TTA. An alternative line of work avoids gradient-based updates entirely. T3A [7] adjusts classifier prototypes using high-confidence pseudo-labels; SHOT [11] aligns target features to fixed source classifiers via information maximization; and AdaNPC [35] replaces the parametric classifier with a k-nearest neighbor mechanism over a memory of source features. These methods offer advantages in efficiency and stability but have been explored only for general visual recognition tasks, leaving their potential for deepfake detection unexplored—a gap our work directly addresses.

3 Problem Statement

We formalize the DF detection task as a binary classification problem and define the test-time adaptation setting under which our method operates.

Source Domain Training. Let $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$ denote the source training dataset, where $x_i^s \in \mathcal{X}$ is a synthetic image and $y_i^s \in \mathcal{Y} = \{0, 1\}$ is its corresponding label, with 0 indicating a pristine image and 1 indicating a manipulated image. A DF detector $f_\theta : \mathcal{X} \to [0, 1]$ parameterized by θ is trained on $\mathcal{D}_s$ drawn from the source distribution $P_s(X, Y)$, such that f_θ learns a mapping that minimizes the expected risk:

$$\theta^* = \arg\min_\theta \, \mathbb{E}_{(x,y)\sim P_s} \left[\mathcal{L}(f_\theta(x), y)\right] \tag{1}$$

where $\mathcal{L}$ denotes the cross-entropy loss. Under the i.i.d. assumption where test data follows the same distribution P_s, the trained detector f_{θ^*} achieves satisfactory performance.

Distribution Shift at Test Time. In real-world deployment, the detector encounters a target domain $\mathcal{D}_t = \{x_j^t\}_{j=1}^{N_t}$ drawn from a shifted distribution $P_t(X, Y) \neq P_s(X, Y)$. This distributional discrepancy arises from two primary sources:

(i) Cross-domain shift. The target data may originate from unseen generative methods, different identity demographics, or novel capture conditions, inducing a covariate shift $P_t(X) \neq P_s(X)$ while the labeling function remains consistent, i.e., $P_t(Y|X) = P_s(Y|X)$.

(ii) Post-processing corruption. Both authentic and manipulated images may undergo transformations $\tau \in \mathcal{T}$ (e.g., JPEG compression, Gaussian blurring, resizing, noise injection) at varying severity levels $l \in \{1, 2, \ldots, L\}$, such that the effective test input becomes $\tilde{x}_j^t = \tau_l(x_j^t)$. These operations degrade the forensic artifacts upon which f_{θ^*} relies, further widening the gap between training and test distributions.

Under either or both of these shifts, the performance of the source-trained detector degrades substantially:

$$\mathbb{E}_{(x,y)\sim P_t} \left[\mathcal{L}(f_{\theta^*}(x), y)\right] \gg \mathbb{E}_{(x,y)\sim P_s} \left[\mathcal{L}(f_{\theta^*}(x), y)\right] \tag{2}$$

Test-Time Adaptation Setting. To address this degradation, we consider the test-time adaptation (TTA) setting subject to the following constraints:

- **(C1)** The source training data $\mathcal{D}_s$ is inaccessible at test time due to privacy or storage limitations.
- **(C2)** No ground-truth labels are available for the target data $\mathcal{D}_t$.
- **(C3)** The model parameters θ^* remain strictly frozen during inference, i.e., no gradient computation or parameter update is performed at test time.

Constraint **(C3)** distinguishes our setting from existing TTA methods for deepfake detection such as $\mathrm{T}^2\mathrm{A}$ [14], which requires solving an online optimization problem $\theta_t \leftarrow \theta^* - \eta \nabla_\theta \mathcal{L}_{\text{aux}}$ at each test step, incurring computational overhead and risking error accumulation under non-stationary streams.

Objective. Under constraints **(C1)**–**(C3)**, we assume access to a small reference set $\mathcal{D}_{\text{ref}} = \{(x_k, y_k)\}_{k=1}^{M}$ comprising K labeled samples per class ($M = NK$, with $N = 2$), which can be drawn from either the source domain or the target domain. Our goal is to construct a non-parametric adaptation mechanism $g : \mathcal{X} \to [0, 1]$ that leverages $\mathcal{D}_{\text{ref}}$ at inference time to reduce the target risk without modifying θ^*:

$$\mathbb{E}_{(x,y)\sim P_t}\left[\mathcal{L}\big(g(x;\, \mathcal{D}_{\text{ref}},\, \theta^*),\, y\big)\right] < \mathbb{E}_{(x,y)\sim P_t}\left[\mathcal{L}(f_{\theta^*}(x),\, y)\right] \tag{3}$$

while preserving the computational efficiency and stability guarantees afforded by a parameter-free inference pipeline.

4 Methodology

We present FreDA, a non-parametric adaptation method that enhances pre-trained deepfake detectors at inference time without modifying any model parameters. Inspired by the work [34], FreDA adapts the framework to the binary classification structure of DF detection by constructing a lightweight key-value cache from a small reference set and combining its predictions with the detector's prior knowledge. The remainder of this section is organized as follows: Section 4.1 describes the construction of the key-value cache model. Section 4.2 details the training-free inference mechanism via cache retrieval. Finally, Section 4.3 introduces an optional cache fine-tuning strategy to further improve detection performance. Fig. 2 illustrates the pipeline of FreDA.

4.1 Cache Model Construction for Deepfake Detection

Adapting large-scale vision-language models to the forensic domain (e.g., CLIP) through conventional linear probing requires substantial labeled data and gradient-based optimization, both of which conflict with the efficiency and privacy constraints outlined in Section 3. We instead propose a *non-parametric* adaptation mechanism that encodes domain-specific knowledge into a fixed lookup structure, eliminating the need for any learnable parameters.

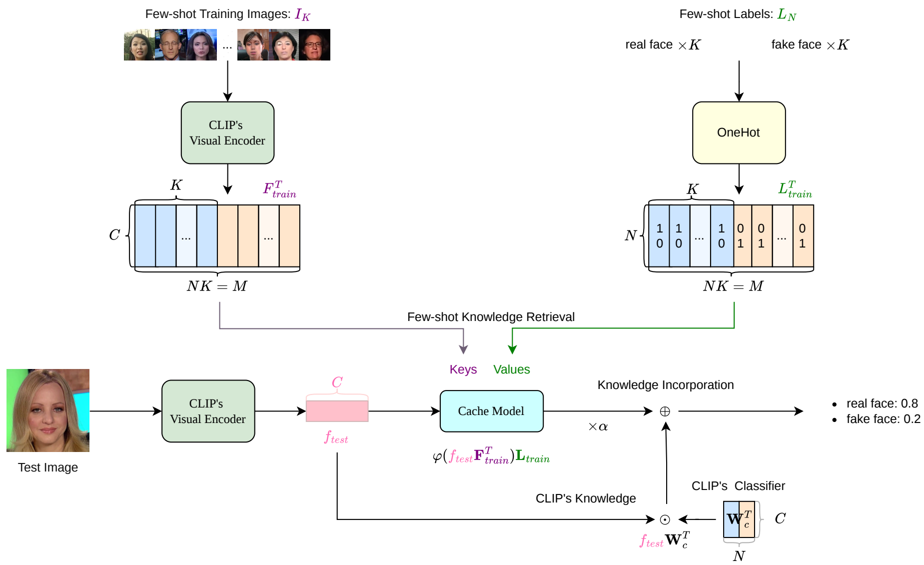

Fig. 2. The pipeline of FreDA. Given a reference set $\mathcal{D}_{\text{ref}}$ of K samples per class, the cache construction phase encodes each image through the frozen CLIP visual encoder and pairs the L_2-normalized features $\mathbf{F}_{\text{ref}}$ with their one-hot label vectors $\mathbf{L}_{\text{ref}}$. At inference time, a test image is encoded into f_{test} and matched against $\mathbf{F}_{\text{ref}}$ via cosine similarity to produce a cache-based prediction. This prediction is then combined with the output of CLIP's textual classifier W_c^T through a residual ratio α to yield the final classification logits.

Concretely, let $\mathcal{D}_{\text{ref}} = \{(x_k, y_k)\}_{k=1}^{M}$ denote the reference set defined in Section 3, comprising K samples per class across $N = 2$ classes ($M = NK$). Each reference image x_k is passed through the frozen CLIP visual encoder to obtain a C-dimensional representation, which is subsequently L_2-normalized so that inner products correspond to cosine similarities. Aggregating these representations row-wise yields the feature matrix $\mathbf{F}_{\text{ref}} \in \mathbb{R}^{M\times C}$. Independently, the binary label of each sample is mapped to a one-hot vector, producing the label matrix $\mathbf{L}_{\text{ref}} \in \mathbb{R}^{M\times N}$:

$$\mathbf{F}_{\text{ref}} = \text{Normalize}\big(\text{VisualEncoder}(\{x_k\}_{k=1}^{M})\big), \tag{4}$$

$$\mathbf{L}_{\text{ref}} = \text{OneHot}(\{y_k\}_{k=1}^{M}). \tag{5}$$

The pair $(\mathbf{F}_{\text{ref}}, \mathbf{L}_{\text{ref}})$ constitutes our cache: $\mathbf{F}_{\text{ref}}$ indexes each reference sample in the shared vision-language embedding space, while $\mathbf{L}_{\text{ref}}$ records its ground-truth class membership. There are two properties of this construction. First, the cache is *entirely deterministic* given $\mathcal{D}_{\text{ref}}$: because the encoder is frozen and the labels are fixed annotations, no optimization objective is involved and the result is reproducible. Second, the binary structure of DF detection ($N = 2$) yields a highly compact label matrix whose columns directly partition the cache into authentic and manipulated subsets, enabling the retrieval mechanism described next to produce a soft real-versus-fake vote with a single matrix product.

4.2 Training-Free Adaptation via Cache Retrieval

At inference time, the detector must classify an unseen test image x_j^t from the target domain $\mathcal{D}_t$ without performing any gradient computation, in accordance with constraint (C3). We achieve this by treating the test sample's representation as a query against the cache constructed in Section 4.1, aggregating the stored labels according to their relevance to the query.

Similarity Computation. The test image is first encoded by the same frozen CLIP visual encoder used during cache construction, producing an L_2-normalized feature vector $f_{\text{test}} \in \mathbb{R}^{1\times C}$. Because both f_{test} and the rows of $\mathbf{F}_{\text{ref}}$ lie on the unit hypersphere, their dot product directly measures cosine proximity. To convert these raw similarities into non-negative relevance weights, we apply an exponential kernel:

$$A = \exp\bigl(-\beta\,(1 - f_{\text{test}}\,\mathbf{F}_{\text{ref}}^{\top})\bigr), \tag{6}$$

where $A \in \mathbb{R}^{1\times M}$ and $\beta > 0$ is a concentration parameter. Intuitively, β governs the locality of the retrieval: large values concentrate weight on the nearest neighbors in the cache, while small values yield a more uniform aggregation. This trade-off is particularly relevant in deepfake detection, where authentic and manipulated faces may be closely interleaved in feature space due to shared identity content.

Prediction via label aggregation. The weighted sum $A\mathbf{L}_{\text{ref}} \in \mathbb{R}^{1\times N}$ produces a soft class distribution derived solely from the cache. To preserve the general visual semantics captured during CLIP's pre-training, we complement this cache-derived signal with the detector's own zero-shot prediction, computed as $f_{\text{test}}\,W_c^{\top}$, where $W_c \in \mathbb{R}^{N\times C}$ encodes the textual class prototypes. The final logits are obtained by a convex-style combination of the two terms:

$$\text{logits} = \alpha\,A\,\mathbf{L}_{\text{ref}} \;+\; f_{\text{test}}\,W_c^{\top}, \tag{7}$$

where $\alpha > 0$ controls how strongly the cache-based evidence modulates the pre-trained output. Under large domain shifts, increasing α allows the reference set to supply corrective distributional information. Conversely, when the target domain is well-aligned with CLIP's pre-training distribution, a smaller α defers to the model's original predictions.

The entire inference path defined by Eqs. (6)–(7) reduces to two matrix: vector products and a scalar-weighted addition, all executed over the frozen encoder output. No backward pass is triggered, and no entry of θ^* is modified, ensuring that adaptation introduces negligible latency and zero risk of error accumulation. These properties are critical for streaming deployment scenarios where the test distribution may shift continuously.

4.3 FreDA-F: Enhancing the Cache via Supervised Refinement

While FreDA achieves competitive adaptation without any parameter learning, its representational capacity is bounded by the quality of the frozen CLIP features stored in the cache. In the DF detection setting, the pre-trained visual encoder was never exposed to forgery-specific cues during its original training, so the cached feature vectors may not optimally separate real and manipulated samples—particularly under aggressive post-processing or when the generative model produces artefacts that are subtle in CLIP's native feature space.

To address this limitation, we introduce **FreDA-F**, a lightweight supervised extension that treats the reference feature matrix $\mathbf{F}_{\text{ref}}$ as an initialization for a small set of trainable parameters, while leaving all other components unchanged. Specifically, the label matrix $\mathbf{L}_{\text{ref}} \in \mathbb{R}^{M\times 2}$, the CLIP visual encoder, and the textual classifier W_c remain frozen. Only the $M \times C$ entries of $\mathbf{F}_{\text{ref}}$ are released for gradient-based optimization, supervised by the cross-entropy loss on the reference set $\mathcal{D}_{\text{ref}}$.

Rationale. Recall from Eq. (6) that the relevance weights A are determined entirely by the cosine similarities between the test feature and the cached reference features. Optimizing $\mathbf{F}_{\text{ref}}$ therefore reshapes the similarity landscape: reference vectors that initially lie in ambiguous regions of the embedding space are pushed toward class-discriminative positions, sharpening the separation between real and fake prototypes. By contrast, the label matrix encodes ground-truth annotations and must remain fixed to preserve the integrity of the class assignments.

Fine-tuning protocol. Because the number of trainable parameters ($M \times C$) is orders of magnitude smaller than the full detector, FreDA-F converges rapidly using standard optimizers such as AdamW with cosine learning-rate scheduling. The starting point provided by CLIP's pre-trained features is already well-structured (as evidenced by FreDA's strong training-free performance), so the optimization performs only a modest refinement rather than learning representations from scratch.

Preserving deployment flexibility. An important design consideration is that FreDA-F's fine-tuning occurs *offline* on the reference set, *not* during test-time inference. Once the refined $\mathbf{F}^*_{\text{ref}}$ is obtained, the inference pipeline remains identical to Eqs. (6)–(7): no gradients are computed on incoming test samples, and the base detector's parameters θ^* are never modified. FreDA-F therefore retains the core advantage of training-free test-time adaptation: zero risk of error accumulation under non-stationary test streams while leveraging a one-time, low-cost optimization to improve the cache's discriminative power.

5 Experiments

5.1 Experimental Setup

Datasets. We conduct experiments on FaceForensics++ (FF++) [24] for in-domain evaluation, and utilize Celeb-DF-v1 [32] and DFDCP [5] to assess cross-dataset generalization.

Baselines. Our evaluation includes Zero-shot CLIP [23], which uses text-based prompts for training-free detection, and Linear Probing [8], which trains a classification head on frozen CLIP features.

Implementation Details. All experiments are implemented in PyTorch and conducted on a single NVIDIA RTX 5090 GPU. All input images are resized to 224×224 pixels and normalized per standard CLIP requirements. The ViT-L/14 backbone remains strictly frozen throughout all feature extraction and adaptation phases.

For our proposed FreDA-F framework, the cache model is constructed using a 16-shot subset. The cache keys are optimized for 40 epochs with a batch size of 256, utilizing the AdamW optimizer with a learning rate of 0.0004 and a cosine annealing scheduler. Early stopping is applied by monitoring the validation accuracy to save the optimal weights. The knowledge retrieval hyperparameters, including the residual ratio α and sharpness factor β, are initialized to 2.5 and 3.0, respectively. To fully unlock the adaptation potential, we conduct a fine-grained hyperparameter search on the validation set to dynamically locate the optimal α and β, utilizing a search scale of [5, 4] with a high-resolution search step of [300, 300]. Crucially, the cross-dataset generalization, robustness tests, and ablation studies strictly follow this exact same configuration. Early stopping is applied by monitoring the validation accuracy to save the optimal weights. For the standard Linear Probing baseline, a classification head is trained on the full-scale dataset using SGD (learning rate 0.01, batch size 128) for 20 epochs.

To guarantee rigorous evaluation, the optimal cache model derived from the FF++ setup is directly deployed for all subsequent tests without any further target-domain parameter updates. Crucially, the aforementioned adaptation configurations are strictly maintained across all cross-dataset, robustness, and ablation studies. For the Robustness evaluation, test images are corrupted using five escalating severity levels of Gaussian Blur (kernel sizes: 5, 9, 13, 17, 21), Resize (pixel resolutions: 128, 85, 64, 51, 41), Color Saturation, and Color Contrast (factors: 1.2, 1.5, 1.8, 2.1, 2.5). Finally, for the Ablation Study, we isolate the impact of the cache capacity by solely varying $K \in \{1, 2, 4, 8\}$, while strictly locking all other optimization variables.

5.2 In-Domain Evaluation

To evaluate the effectiveness of our proposed method in an in-domain setting, we conduct experiments on the FaceForensics++ (FF++) dataset. We integrate

FreDA and its fine-tuned variant, FreDA-F, into two distinct baseline frameworks: Zero-shot CLIP and Linear Probing. The quantitative results, including Accuracy (Acc), F1 score, AUC, and Average Precision (AP), are summarized in Table 1.

Performance on Zero-shot CLIP The Zero-shot CLIP backbone provides a baseline Accuracy of 0.622 and an AUC of 0.593. By simply incorporating the training-free FreDA with a 16-shot cache, we observe a steady improvement in nearly all metrics, particularly with the AUC rising to 0.687 and AP to 0.829. Furthermore, the fine-tuned FreDA-F significantly pushes the performance boundary, achieving an Accuracy of 0.767 and an AUC of 0.848. This demonstrates that even without initial domain-specific training, our few-shot adaptation mechanism can effectively bridge the gap for CLIP-based detectors.

Performance on Linear Probing When applied to the Linear Probing baseline, which is already supervised on the source domain, FreDA continues to show its robustness. The baseline Linear Probing yields an Accuracy of 0.674 but suffers from a relatively low F1 score (0.429), indicating a bias in the decision boundary. After adding FreDA, the F1 score dramatically improves to 0.760, and Accuracy reaches 0.721. The FreDA-F variant further optimizes these results to 0.742 Accuracy and 0.804 F1 score, suggesting that the residual cache connection helps regularize the linear head and provides a more balanced classification.

Table 1. Quantitative results on the FF++ dataset. Performance comparison of the training-free FreDA and the fine-tuned FreDA-F using the vanilla CLIP (ViT-L/14) as the backbone. The best results are highlighted in **bold**.

Method	Acc	F1 score	AUC	AP
Zero-shot CLIP	0.622	0.732	0.593	0.748
Zero-shot CLIP + FreDA	0.645	0.732	0.687	0.829
Zero-shot CLIP + FreDA-F	**0.767**	**0.818**	**0.848**	**0.926**
Linear Probing	0.674	0.429	0.772	0.881
Linear Probing + FreDA	0.721	0.760	0.820	**0.913**
Linear Probing + FreDA-F	**0.742**	**0.804**	**0.820**	0.912

5.3 Cross-Dataset Generalization

Real-world deepfake detection heavily relies on a model's robustness against unseen manipulation techniques and unknown generative algorithms. To rigorously evaluate the out-of-distribution (OOD) generalization capability of our framework, we conduct cross-dataset evaluations. Specifically, as demonstrated in Table 2, we compare the training-free FreDA and its fine-tuned variant, FreDA-F, on two baselines across two challenging datasets: Celeb-DF-v1 [32] and DFDCP [5]. The results confirm that FreDA provides a significant boost to both Zero-shot CLIP and Linear Probing baselines, proving that incorporating a small set of target-domain reference features can calibrate the detector's decision boundary for previously unseen manipulation techniques.

Furthermore, the experiments highlight the superior adaptability of the fine-tuned variant, FreDA-F, which achieves the highest metrics across all OOD benchmarks. For instance, the dramatic improvement of the Linear Probing baseline on Celeb-DF-v1—increasing AUC from 0.583 to 0.913—underscores that our method is not merely a marginal enhancer but a robust solution for cross-dataset transfer. Ultimately, these evaluations establish FreDA as a parameter-efficient framework capable of rescuing pre-trained detectors from performance collapse in diverse, real-world deepfake scenarios.

Table 2. Cross-Dataset Generalization from FF++ to unseen target datasets: Celeb-DF-v1 and DFDCP.

Target Dataset	Method	Acc	F1 score	AUC	AP
Celeb-DF-v1	Zero-shot CLIP	0.518	0.589	0.520	0.641
	Zero-shot CLIP + FreDA	0.604	0.656	0.658	0.781
	Zero-shot CLIP + FreDA-F	**0.626**	**0.658**	**0.709**	**0.825**
	Linear Probing	0.530	0.541	0.583	0.660
	Linear Probing + FreDA	0.554	0.559	0.614	0.693
	Linear Probing + FreDA-F	**0.841**	**0.868**	**0.913**	**0.935**
DFDCP	Zero-shot CLIP	0.520	**0.557**	0.572	0.717
	Zero-shot CLIP + FreDA	0.447	0.325	0.638	0.775
	Zero-shot CLIP + FreDA-F	**0.563**	0.550	**0.709**	**0.835**
	Linear Probing	0.520	0.508	0.607	0.771
	Linear Probing + FreDA	0.507	0.474	0.615	0.775
	Linear Probing + FreDA-F	**0.626**	**0.651**	**0.753**	**0.857**

5.4 Robustness to Post-Processing

In real-world scenarios, digital media are frequently subjected to various post-processing operations during transmission and storage, such as social media compression, resizing, and filtering. These operations often obscure subtle forgery artifacts, posing a significant challenge to the robustness of DF detectors. To evaluate resilience against real-world post-processing, we apply four common perturbations to the FF++ test set—Color Contrast, Color Saturation, Resize, and Gaussian Blur—each at five increasing intensity levels. The results are presented in Fig. 3.

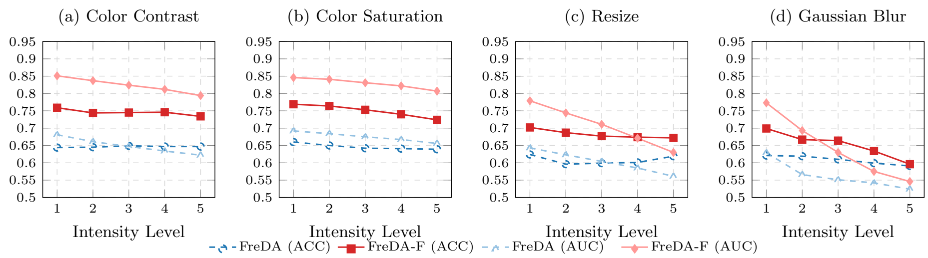

Fig. 3. Robustness to post-processing perturbations on FF++. ACC and AUC of FreDA and FreDA-F under increasing intensity levels of four perturbation types.

FreDA-F consistently outperforms FreDA across all perturbation types and intensity levels. A clear dichotomy emerges between perturbation categories. Under color-based perturbations (Fig. 3a–b), both methods exhibit remarkable stability, with FreDA-F maintaining ACC above 0.72 and AUC above 0.79 even at Level 5, indicating that the frozen CLIP backbone provides representations largely invariant to global pixel-value shifts. In contrast, structural perturbations—Resize and Gaussian Blur (Fig. 3c–d)—induce a steeper decay, as these operations act as low-pass filters that eliminate the high-frequency details essential for detecting localized manipulation artifacts. Despite this more pronounced degradation, FreDA-F consistently preserves a substantial margin over FreDA, confirming the value of cache key refinement for maintaining robustness under challenging conditions.

5.5 Ablation Study

To further explore the sensitivity of our proposed FreDA-F to the available supervision in the target domain, we conduct an ablation study by varying the number of shots in the cache retrieval mechanism. The results are reported in Table 3. Our 16 shots achieved the best performance.

Table 3. Impact of the number of shots. Performance of FreDA-F on FF++ with different shot settings.

Setting	**Acc**	**F1**	**AUC**	**AP**
1-shot	0.765	**0.819**	0.845	0.925
2-shots	0.763	0.817	0.845	0.925
4-shots	0.760	0.813	0.843	0.924
8-shots	0.760	0.812	0.845	0.925
16-shots (**Ours**)	**0.767**	0.818	**0.848**	**0.926**

6 Conclusion

In this work, we introduced FreDA, a novel training-free test-time adaptation method designed to enhance the robustness and generalization of deepfake detectors. By leveraging a non-parametric cache retrieval mechanism and a residual connection strategy, FreDA effectively incorporates distributional information from a small reference set without requiring any gradient computation or parameter updates at inference time. Our extensive evaluations on Celeb-DF-v1 and DFDCP demonstrate that FreDA not only achieves competitive in-domain performance but also exhibits superior transferability and resilience against common post-processing perturbations compared to traditional fine-tuning paradigms. Furthermore, the fine-tuned variant, FreDA-F, provides a flexible trade-off between computational efficiency and detection accuracy. However, in this work, the scalability of cache retrieval mechanism for large reference sets is not fully addressed. The future work should explore more efficient retrieval strategies in FreDA to improve the scalability and performance.

References

1. Cao, J., Ma, C., Yao, T., Chen, S., Ding, S., Yang, X.: End-to-end reconstruction-classification learning for face forgery detection. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 4113–4122 (2022)
2. Chen, L., Zhang, Y., Song, Y., Wang, J., Liu, L.: Ost: Improving generalization of deepfake detection via one-shot test-time training. Advances in Neural Information Processing Systems **35**, 24597–24610 (2022), `https://proceedings.neurips.cc/paper_files/paper/2022/file/9bf0810a4a1597a36d27ceea58667d92-Paper-Conference.pdf`
3. CNN: Finance worker pays out $25 million after video call with deepfake CFO. `https://edition.cnn.com/2024/02/04/asia/deepfake-cfo-scam-hong-kong-intl-hnk/index.html` (2024), accessed: 28 June 2025
4. Corvi, R., Cozzolino, D., Poggi, G., Nagano, K., Verdoliva, L.: Intriguing properties of synthetic images: from generative adversarial networks to diffusion models. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 973–982 (2023), `https://openaccess.thecvf.com/content/CVPR2023W/WMF/html/Corvi_Intriguing_Properties_of_Synthetic_Images_From_Generative_Adversarial_Networks_to_CVPRW_2023_paper.html`
5. Dolhansky, B., Bitton, J., Pflaum, B., Lu, J., Howes, R., Wang, M., Ferrer, C.C.: The deepfake detection challenge (dfdc) dataset (2020), `https://arxiv.org/abs/2006.07397`
6. Frank, J., Eisenhofer, T., Schönherr, L., Fischer, A., Kolossa, D., Holz, T.: Leveraging frequency analysis for deep fake image recognition. In: International conference on machine learning. pp. 3247–3258. PMLR (2020)
7. Iwasawa, Y., Matsuo, Y.: Test-time classifier adjustment module for model-agnostic domain generalization. Advances in Neural Information Processing Systems **34**, 2427–2440 (2021), `https://proceedings.neurips.cc/paper_files/paper/2021/file/1415fe9fea0fa1e45dddcff5682239a0-Paper.pdf`
8. Khan, S.A., Dang-Nguyen, D.T.: Clipping the deception: Adapting vision-language models for universal deepfake detection. In: Proceedings of the

2024 International Conference on Multimedia Retrieval. pp. 1006–1015 (2024). https://doi.org/10.1145/3652583.3658035
9. Layton, S., Tucker, T., Olszewski, D., Warren, K., Butler, K., Traynor, P.: {SoK}: The good, the bad, and the unbalanced: Measuring structural limitations of deepfake media datasets. In: 33rd USENIX Security Symposium (USENIX Security 24). pp. 1027–1044 (2024)
10. Liang, J., He, R., Tan, T.: A comprehensive survey on test-time adaptation under distribution shifts. International Journal of Computer Vision **133**(1), 31–64 (2025)
11. Liang, J., Hu, D., Feng, J.: Do we really need to access the source data? source hypothesis transfer for unsupervised domain adaptation. In: International conference on machine learning. pp. 6028–6039. PMLR (2020), https://proceedings.mlr.press/v119/liang20a.html
12. Liu, H., Li, X., Zhou, W., Chen, Y., He, Y., Xue, H., Zhang, W., Yu, N.: Spatial-phase shallow learning: rethinking face forgery detection in frequency domain. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 772–781 (2021)
13. Nguyen, D., Mejri, N., Singh, I.P., Kuleshova, P., Astrid, M., Kacem, A., Ghorbel, E., Aouada, D.: Laa-net: Localized artifact attention network for quality-agnostic and generalizable deepfake detection. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 17395–17405 (2024)
14. Nguyen-Le, H.H., Tran, V.T., Nguyen, D.T., Le-Khac, N.A.: Think twice before adaptation: improving adaptability of deepfake detection via online test-time adaptation. In: Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence. pp. 7679–7687 (2025)
15. Nguyen-Le, H.H., Tran, V.T., Nguyen, T., Le-Khac, N.A.: A survey on proactive deepfake defense: Disruption and watermarking. ACM Comput. Surv. **58**(5) (Nov 2025). https://doi.org/10.1145/3771296
16. Ni, Y., Meng, D., Yu, C., Quan, C., Ren, D., Zhao, Y.: Core: Consistent representation learning for face forgery detection. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 12–21 (2022)
17. Niu, S., Wu, J., Zhang, Y., Chen, Y., Zheng, S., Zhao, P., Tan, M.: Efficient test-time model adaptation without forgetting. In: International conference on machine learning. pp. 16888–16905. PMLR (2022), https://proceedings.mlr.press/v162/niu22a.html
18. Niu, S., Wu, J., Zhang, Y., Wen, Z., Chen, Y., Zhao, P., Tan, M.: Towards stable test-time adaptation in dynamic wild world. In: The Eleventh International Conference on Learning Representations (2023), https://doi.org/10.48550/arXiv.2302.12400
19. Ojha, U., Li, Y., Lee, Y.J.: Towards universal fake image detectors that generalize across generative models. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 24480–24489 (2023)
20. Pan, K., Yin, Y., Wei, Y., Lin, F., Ba, Z., Liu, Z., Wang, Z., Cavallaro, L., Ren, K.: Dfil: Deepfake incremental learning by exploiting domain-invariant forgery clues. In: Proceedings of the 31st ACM International Conference on Multimedia. pp. 8035–8046 (2023). https://doi.org/10.1145/3581783.3612377
21. Pearson, J., Zinets, N.: Deepfake footage purports to show ukrainian president capitulating. https://www.reuters.com/world/europe/deepfake-footage-purports-show-ukrainian-president-capitulating-2022-03-16 (2022), accessed: 12 February 2026

22. Qian, Y., Yin, G., Sheng, L., Chen, Z., Shao, J.: Thinking in frequency: Face forgery detection by mining frequency-aware clues. In: European conference on computer vision. pp. 86–103. Springer (2020)
23. Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., Sutskever, I.: Learning transferable visual models from natural language supervision. In: Meila, M., Zhang, T. (eds.) Proceedings of the 38th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 139, pp. 8748–8763. PMLR (18–24 Jul 2021)
24. Rossler, A., Cozzolino, D., Verdoliva, L., Riess, C., Thies, J., Niessner, M.: Faceforensics++: Learning to detect manipulated facial images. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) (October 2019)
25. Rössler, A., Cozzolino, D., Verdoliva, L., Riess, C., Thies, J., Nießner, M.: Faceforensics: A large-scale video dataset for forgery detection in human faces (2018)
26. Shiohara, K., Yamasaki, T.: Detecting deepfakes with self-blended images. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 18720–18729 (2022)
27. Sun, Y., Wang, X., Liu, Z., Miller, J., Efros, A., Hardt, M.: Test-time training with self-supervision for generalization under distribution shifts. In: International conference on machine learning. pp. 9229–9248. PMLR (2020)
28. Wang, D., Shelhamer, E., Liu, S., Olshausen, B., Darrell, T.: Tent: Fully test-time adaptation by entropy minimization (2021)
29. Wang, D., Shelhamer, E., Liu, S., Olshausen, B., Darrell, T.: Tent: Fully test-time adaptation by entropy minimization. In: International Conference on Learning Representations (2021)
30. Wang, Q., Fink, O., Van Gool, L., Dai, D.: Continual test-time domain adaptation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 7201–7211 (2022)
31. Wang, Z., Bao, J., Zhou, W., Wang, W., Li, H.: Altfreezing for more general video face forgery detection. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 4129–4138 (2023)
32. Yuezun Li, Xin Yang, P.S.H.Q., Lyu, S.: Celeb-df: A large-scale challenging dataset for deepfake forensics. In: IEEE Conference on Computer Vision and Patten Recognition (CVPR) (2020)
33. Zhang, M., Levine, S., Finn, C.: Memo: Test time robustness via adaptation and augmentation. Advances in neural information processing systems **35**, 38629–38642 (2022)
34. Zhang, R., Zhang, W., Fang, R., Gao, P., Li, K., Dai, J., Qiao, Y., Li, H.: Tip-adapter: Training-free adaption of clip for few-shot classification. In: European conference on computer vision. pp. 493–510. Springer (2022)
35. Zhang, Y., Wang, X., Jin, K., Yuan, K., Zhang, Z., Wang, L., Jin, R., Tan, T.: Adanpc: Exploring non-parametric classifier for test-time adaptation. In: International conference on machine learning. pp. 41647–41676. PMLR (2023)
36. Zhao, H., Liu, Y., Alahi, A., Lin, T.: On pitfalls of test-time adaptation. In: Proceedings of the 40th International Conference on Machine Learning. pp. 42058–42080 (2023)